Assignment 1

Part 1

1. DBMS reduces data redundancy and supports concurrent access while in traditional file systems, we are unable to do it.
   In movie ticket booking, DBMS safely handles multiple users trying to book the exact same seat at the same time, whereas a file system might accidentally let both book it
2. A weak entity is drawn with a double rectangle, while a strong entity is a single rectangle. Primary key of weak identity can be formed by Identifying strong entity's primary key + partial key. This is called identifying.
3. Multivalued since a student may have multiple phone numbers.
4. A candidate key is a specific type of super key that cannot be reduced further.
   Candidate keys: roll_no, email
   Super keys: {name, roll_no}, {email, branch}
5. Constraint is foreign key. It protects against orphaned records. It will reject insert operation.
6. Junction Table must be created. Primary key consists of combined foreign keys of the participating entities. It would violate First Normal Form (1NF) by requiring multi-valued cells.

Part 2

Part A

Entities: Customer, Restaurant, Order, Menu Item.

Customer-Order: 1:N. A customer can place multiple orders.
Order-Restaurant: N:1. Each order is associated with exactly one restaurant.
Order-Menu Item: M:N. An order contains multiple menu items, and a menu item can exist in multiple orders.
Restaurant-Menu Item: 1:N. A restaurant can offer multiple menu items.

Part B

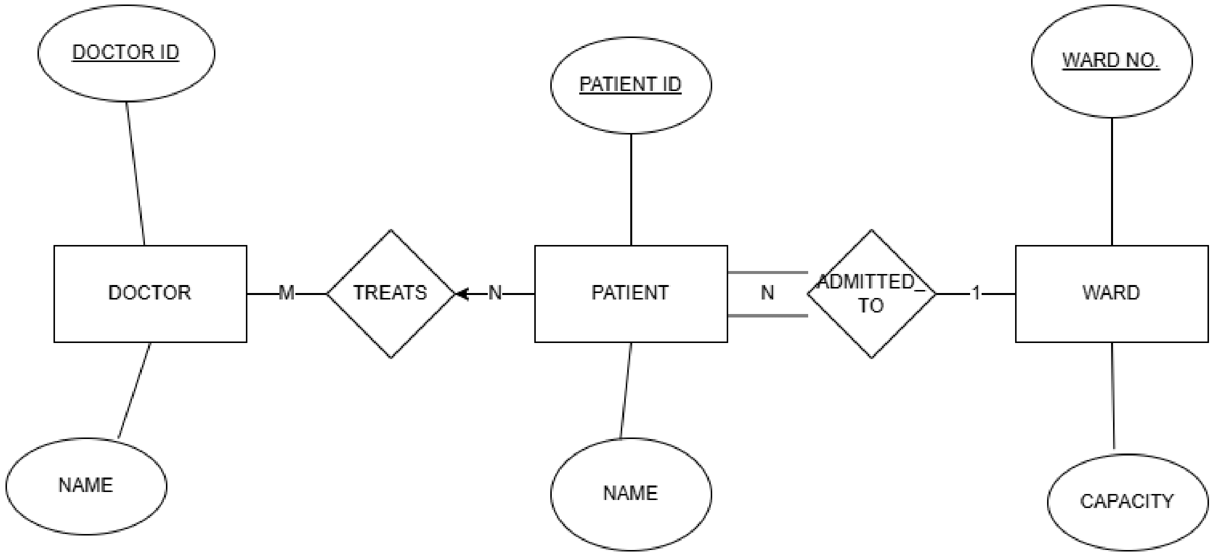
Multivalued: Phone_number. A customer or restaurant can have multiple contact numbers.

Composite: Address. Can be subdivided into smaller parts like street, city, and ZIP code.

Part C

OrderItem (order_id, item_id, quantity)

Part 3



Doctor-Patient M:N: Required because a doctor can treat multiple patients, and a patient can be treated by multiple doctors.

- **Associative Entity:** Treatment.
- **Composite PK:** {doctor_id, patient_id}.

Participation Constraints:

- **Patient in Patient-Ward:** Total. Every patient admitted to the hospital must be assigned a ward.
- **Doctor in Doctor-Patient:** Partial. A doctor exists in the system even if they are not currently treating any admitted patients.

Attribute Design Decision: Patient_Name is modeled as a composite attribute (first_name, last_name) to easily allow sorting, filtering, and querying by surname.